spacing is $a \sim 1$ Å. Thus we find that

$$\Delta k \sim \frac{10^{-4}}{1 \text{ Å}}$$

is the range of wave numbers contained in the wave function for a Cooper pair. A very general property of waves ((3-14), which leads to the uncertainty principle) then immediately tells us that the extent in space of the wave function is

$$\Delta x \sim \frac{1}{\Delta k} \sim 10^4 \text{ Å}$$

This is the size of a typical Cooper pair.

(b) Estimate the density of Cooper pairs in a superconductor.

Example 13-1 shows that the density of conduction electrons in a metal is $n \sim 10^{22}/\text{cm}^3$. The fraction that will form Cooper pairs in a superconductor is of the order of $\Delta k/k_F \sim 10^{-4}$. So

$$n_{\text{Cooper pairs}} \sim 10^{18}/\text{cm}^3$$

Note that the volume of one pair is $\sim(10^4 \text{ Å})^3 = (10^{-4} \text{ cm})^3 = 10^{-12} \text{ cm}^3$. So each such volume contains $\sim 10^6$ overlapping pairs! ◀

The width of the forbidden gap, and the density of quantum states, in a superconductor can be determined from the current-voltage characteristic of a tunnel junction. In such junctions a thin oxide layer ($\sim 10^{-9}$ m thick) separates a normal and a superconducting metal. Electrons tunnel through the barrier, which the nonconducting oxide layer represents, with the aid of an applied voltage. In 1962, Josephson predicted that if the metals on both sides of the junction are superconducting, a current can flow when no voltage is supplied. If a small voltage ($\sim$ a few millivolts) is applied, an alternating current of frequency in the microwave range results. These effects can be used to detect extremely small voltage differences and to measure with enormous precision the ratio e/h used in determination of the fundamental physical constants. Other superconducting effects predicted by Josephson permit a number of quantum properties to be seen in a very simple way, particularly the quantization of magnetic flux, discussed below.

There are many important applications of superconductivity. An obvious application is to superconducting electromagnets, whose fields arise from resistanceless currents flowing through the magnet windings, for use in electric motors and generators. A difficulty is that magnetic fields tend to be induced in the wires of the windings, which tends to destroy their superconductivity. But progress is being made in finding what are called *Type II superconductors*, which have Cooper pairs whose dimensions are small enough to allow a magnetic field to thread its way through the length of a wire in a set of localized channels. These channels lose their superconductivity, but the channels in between them do not. Several niobium-titanium alloys have been found which are Type II superconductors, and they also have the convenience of relatively high critical temperatures ($T_c \simeq 20°\text{K}$).

The absence of power dissipation in superconducting elements makes possible many electronic applications in which space requirements and transmission time requirements are limited, as in computers. Because superconductors are diamagnetic, they can be used to shield out unwanted magnetic flux. This can be put to use in shaping the magnetic lens system of an electron microscope, for example, to eliminate stray field lines and to greatly improve the practical resolving power of the instrument thereby.

Apart from such technological applications of superconductivity, of which a great many more can be cited, there is an increasing application of the theoretical ideas to other fields of physics. For example, these ideas have been applied to analyzing